Weka:- Weka is a collection of machine learning Algorithms for performing the data mining task.

* This is a WEKA GUI Chooser (Graphical User Interface)
* To perform applications then click Explorer button. and the new window is WEKA Explorer. Here different data mining technbluesquelues that are
  preprocess, classify, cluster and associate.

Data Preprocessing:- The process of transforming row data into an understandable format

* WEKA tool uses arff files (Attribute Relation File Format)
* Arff files can be classified into two types
  (i) header information
  (ii) data information

Dataset:- A dataset is a collection of related discrete items of related data that may be accessed individually.

Association:- Associate is a approri algorithm its minimum count value is greater than transaction count value.

Clustering:- It is a simple K-Means Algorithm.

Classification:- Classification is a data mining function that assigns in a collection of target classes.

Experiment-1

Aim :- Demonstration of preprocessing on Dataset student.arff

Filename :- Student.arff

```
@relation student
@attribute age{<30,'30-40',>40}
@attribute income{high,medium,low}
@attribute student{no,yes}
@attribute 'credit-rating'{fair,excellent}
@attribute buyspc{no,yes}
@data
<30, high, no, fair, no
<30, high, no, Excellent, no
30-40, high, no, fair, yes
>40, low, medium, no, fair, yes
>40, low, yes, fair, yes
>40, low, yes, Excellent, yes
<30, medium, no, fair, no
<30, low, yes, fair, no
>40, medium, yes, Excellent, yes
<30, medium, yes, Excellent, yes
30-40, medium, no, Excellent, yes
30-40, high, yes, fair, yes
>40, medium, no, Excellent, no
```

Output:-

Handling of Missing values:

(i) Remove missing values

choose Tab → Weka → Filters → Unsupervised → Instances → Remove with values

(ii) set index values of attribute

1. set Match missing values "True".
2. Replacing Missing values

Unsupervised → attribute ⇒ replace missing values

* Data type conversion

(i) Supervised → attribute → discretise datatype change

(ii) Unsupervised → attribute → discretise → gives range

1. set index of attribute and select bins
2. set precision. set equal to bin frequency
3. set to True.

* Attribute Selection

(i) CFS subset Eval - best fit

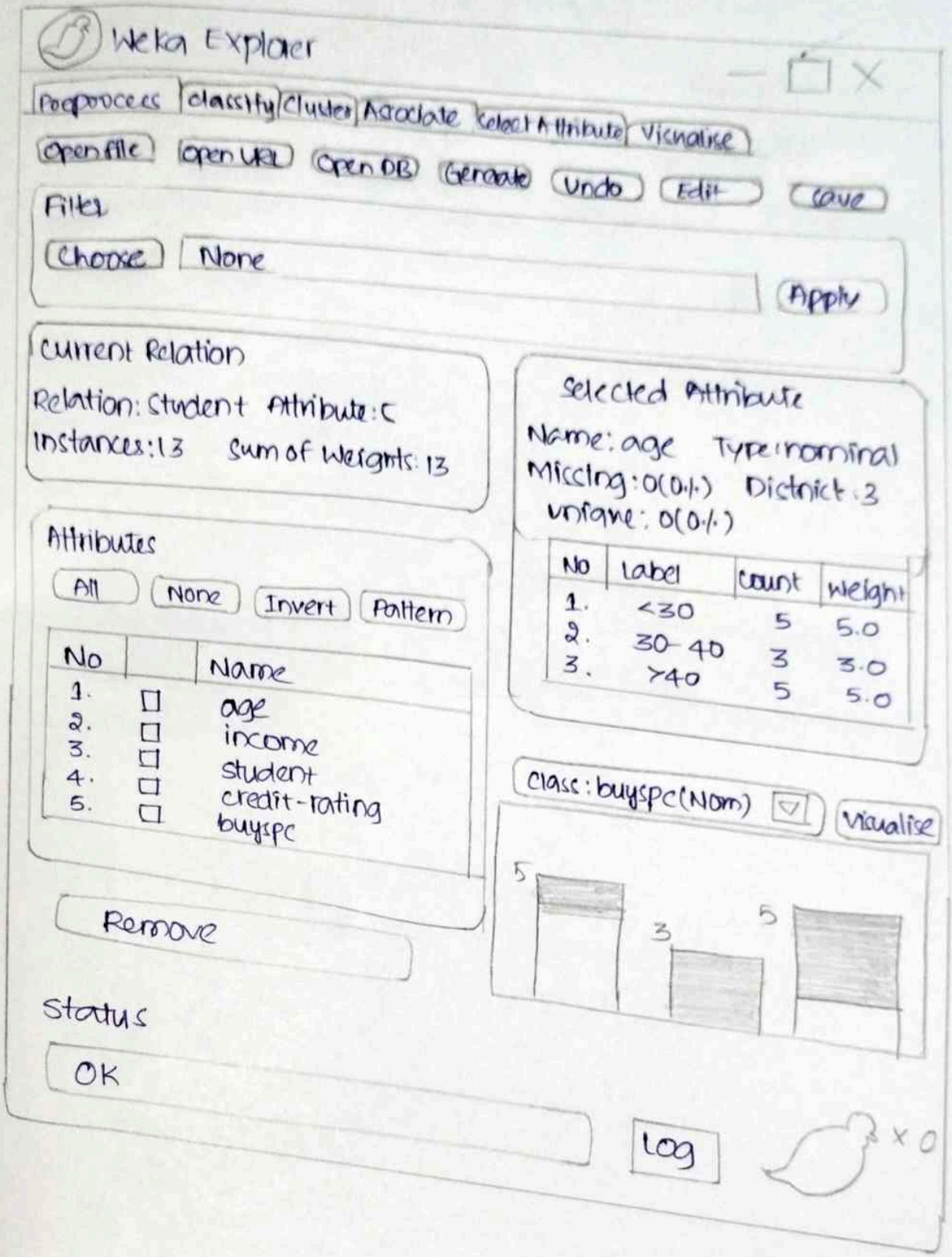
(ii) Information again attribute Eval Ranker.

Normalisation:- Normalisation is a process of changing numeric values to predefined values of 0 to 1. (or conversion of larger numeric values into 0's and 1's ranges.

To click unsupervised → attribute → normalise

Dimensionality reduction :- The data never be lost. There is no loss in data. It reduces the large data into small data.

Unsupervised → attribute → principle component.



Appriori Algorithm:- Appriori algorithm is used for finding frequent item set in the given transaction in database

* The items are that are appear frequently

Itemset:- Itemset is a set of items put in the transaction are called Itemset.

Ex:- Let us consider the transaction database D as shown below there are 9 transactions in the database use appriori algorithm for finding frequent itemsets in database.

Assume that min support count value = 2.

Tid	List Of Items
T1	I1, I2, I5
T2	I2, I4
T3	I2, I3
T4	I1, I2, I4
T5	I1, I3
T6	I2, I3
T7	I1, I3
T8	I1, I2, I3, I5
T9	I1, I2, I3

Step-1:- scan D(data) for count of each candidate

C1(Candidate Support Count)

Items	Support Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

Compare cand support count with minimum support count →

L1 (Watt/Results)

Items	Support Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

(i) support count is less than minimum support count i.e. taken from the database

(ii) min support count is greater than support count that exists in candidate item set

step-2: Generate C2 from L1

C2

Items	Support Count
{I1, I2}	4
{I1, I3}	4
{I1, I4}	1 ⊗
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0 ⊗
{I3, I5}	1 ⊗
{I4, I5}	0 ⊗

Compare cand supp count with min support count →

L2

Items	Support Count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

step-3:- Generate C3 from L2

C3

Items	Support Count
{I1, I2, I3}	2
{I1, I2, I5}	2
{I1, I2, I4}	1 ⊗
{I1, I3, I5}	1 ⊗
{I2, I3, I4}	0 ⊗
{I2, I3, I5}	1 ⊗
{I2, I4, I5}	0 ⊗

Compare cand supp count with minimum support count →

L3

Items	Support Count
{I1, I2, I3}	2
{I1, I2, I5}	2

Step-4: Generate from L3 to C4

Items	Support count
{I1, I2, I3, I4, I5}	1 ⊗

→

Items	Support count
ϕ	ϕ

Note: Whenever $L4 = \phi$ the algorithm terminated the item sets in L3 (or) frequent i.e {I1, I2, I3} & {I1, I2, I5}

This algorithm is terminated $Lk = \phi$ ie $L4 = \phi$

$Lk-1$ i.e $L4-1 = L3$

Experiment-2

Aim:- Demonstration of association rule of process of dataset contactlenses.arff using apriori algorithm

```
@relation contact-lenses
@attribute age {young, pre-presbyopic, presbyopic}
@attribute spectacle-prescription {myope, hypermetrope}
@attribute astigmatism {no, yes}
@attribute tear-prod-rate {reduced, normal}
@attribute contact-lenses {soft, hard, none}
@data
%
% 24 instances
%
young, myope, no, reduced, none
young, myope, no, normal, soft
young, myope, yes, reduced, none
young, myope, yes, normal, hard
young, hypermetrope, no, reduced, none
young, hypermetrope, no, normal, soft
young, hypermetrope, yes, reduced, none
young, hypermetrope, yes, normal, hard
pre-presbyopic, myope, no, reduced, none
pre-presbyopic, myope, no, normal, soft
pre-presbyopic, myope, yes, reduced, none
pre-presbyopic, myope, yes, normal, hard
pre-presbyopic, hypermetrope, no, reduced, none
pre-presbyopic, hypermetrope, no, normal, soft
pre-presbyopic, hypermetrope, yes, reduced, none
pre-presbyopic, hypermetrope, yes, normal, none
pre-presbyopic, myope, no, reduced, none
pre-presbyopic, myope, no, normal, none
pre-presbyopic, myope, no, reduced, none
pre-presbyopic, myope, no, normal, hard
```

presbyopic, hypermetrope, no, reduced, none
presbyopic, hypermetrope, no, normal, soft
presbyopic, hypermetrope, yes, reduced, none
presbyopic, hypermetrope, yes, normal, none
%
%

Output:-

Associate → choose → Apriori

set lowerBoundMinSupport = 0.25
set upperBoundMinSupport = 0.9.

Apriori

Minimum support: 0.25
Minimum metric <confidence>: 0.9
Number of cycles performed: 15

Generated sets of large itemsets:

Size of set of large itemsets L(1): 10
size of set of large itemsets L(2): 18
Size of set of large itemsets L(3): 4

Best rules found:

1. tear-prod-rate = reduced 12 ==> contact-lenses = none 12 <conf: (1) > lift: (1.6) lev: (0.19) [4] conv: (4.5)
2. spectacle-prescrip = myope tear-prod-rate 6 ==> contact-lenses = none 6 <conf: (1) > lift: (1.6) lev: (0.09) [2] conv: (2.25)
3. spectacle-prescrip = hypermetrope tear-prod-rate-reduced 6 ==> contact-lenses = none 6 <conf: (1) > lift: (1.6) lev: (0.09) [2] conv: (2.25)

14. astigmatism=no tear-prod-rate=reduced 6 ==> contact-lenses
=none 6 <conf:(1)> lift:(1.6) lev:(0.09) [2] conv:(2.26)

15. astigmatism=yes tear-prod-rate=reduced 6 ==> contact-lenses
=none 6 <conf:(1)> lift: (1.6) lev:(0.09) [2] conv:(2.26)